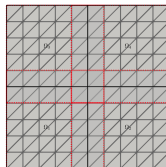
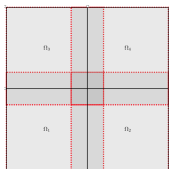


# *A posteriori* error analysis of overlapping domain decomposition techniques

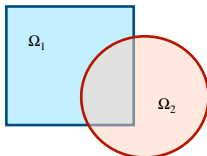
Jehanzeb H. Chaudhry\*, Simon Tavener, Don Estep

\* Department of Mathematics and Statistics  
University of New Mexico

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University of Nevada, Las Vegas (2019)



# Overlapping Additive and Multiplicative Schwarz



## Additive or Jacobi

Solve on  $\Omega_1$  and  $\Omega_2$  independently

Use old solution from  
other domain on the overlap

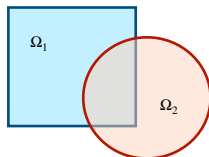
## Multiplicative or Gauss-Seidel

Solve on  $\Omega_1$  and  $\Omega_2$  alternately

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Schwarz methods are promising for next generation HPC architectures.

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Schwarz methods are promising for next generation HPC architectures.

This talk: Error in a quantity-of-interest computed using discretized Schwarz.

## *A posteriori* analysis in linear algebra

Let  $U$  be an approximate solution of

$$Au = b.$$

We seek to compute a quantity of interest given by a linear functional

$$(\psi, u) = \psi^T u.$$

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The **error**

$$e = u - U$$

is *not* computable, but we can compute the residual  $R$ ,

$$R = b - \mathbf{A}U = \mathbf{A}u - \mathbf{A}U = \mathbf{A}e.$$

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The *computable* **error representation** is

$$(\psi, u) - (\psi, U) = (\psi, e) = (\mathbf{A}^\top \phi, e) = (\phi, \mathbf{A}e) = (\phi, R).$$

## Previous work on adjoint-based error estimation

1. Multirate time integration [Chaudhry, Estep, Ginting, Tavener. 2013]
2. Iterative methods for ODEs [Chaudhry, Estep, Ginting, Tavener. 2015]
3. Implicit-Explicit time integration methods [Chaudhry, Estep, Ginting, Tavener. 2015], [Chaudhry, Collins, Shadid. 2017]
4. Continuum biomolecular models (PBE) [Chaudhry. 18]
5. Parallel-in-time integration [Chaudhry, Estep, Tavener, Carey, Sandelin. 2016]
6. Lions Non-overlapping Domain decomposition [Chaudhry, Burch, Estep. 2018]
7. **Overlapping Domain decomposition**

**Key: Identify distinct sources of error**

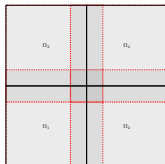
# Overlapping multiplicative Schwarz

Find  $u \in H_0^1(\Omega)$  such that

$$a(u, v) = l(v) \quad \forall v \in H_0^1(\Omega),$$

where  $a(\cdot, \cdot)$  is a bilinear form over  $\Omega$ .

Assume we have  $p$  overlapping subdomains  $\Omega_1, \dots, \Omega_p$ .



Let  $a_i(\cdot, \cdot)$  be the restriction of  $a(\cdot, \cdot)$  over  $\Omega_i$  and  $a_{ij}(\cdot, \cdot)$  its restriction over  $\Omega_i \cap \Omega_j$ .

Let  $(\cdot, \cdot)_i$  and  $(\cdot, \cdot)_{ij}$  represent the  $L_2(\Omega_i)$  and  $L_2(\Omega_i \cap \Omega_j)$  inner products respectively.

---

**Algorithm 1** Overlapping multiplicative Schwarz domain decomposition

---

Given  $u^{\{0\}}$  defined on  $\Omega$

**for**  $k = 0, 1, 2, \dots, K - 1$  **do**

**for**  $i = 1, 2, \dots, p$  **do**

        Find  $\tilde{u}^{\{k+i/p\}} \in H_{D_k}^1(\Omega_i)$  such that

$$a_i(\tilde{u}^{\{k+i/p\}}, v) = l_i(v), \quad \forall v \in H_0^1(\Omega_i). \quad (1)$$

        Let

$$u^{\{k+i/p\}} = \begin{cases} \tilde{u}^{\{k+i/p\}}, & \text{on } \overline{\Omega}_i, \\ u^{\{k+(i-1)/p\}}, & \text{on } \Omega \setminus \overline{\Omega}_i. \end{cases} \quad (2)$$

**end for**

**end for**

---

Here  $H_{D_k}^1(\Omega_i) \equiv \{v \in H^1(\Omega_i) \mid v = u^{\{k+(i-1)/p\}} \text{ on } \partial\Omega_i\}$ .

## Reformulated overlapping multiplicative Schwarz

Define  $w^{\{k+i/p\}} = \left( \tilde{u}^{\{k+i/p\}} - u^{\{k+(i-1)/p\}} \right) \in H_0^1(\Omega_1)$

# Reformulated overlapping multiplicative Schwarz

Define  $w^{\{k+i/p\}} = \left( \tilde{u}^{\{k+i/p\}} - u^{\{k+(i-1)/p\}} \right) \in H_0^1(\Omega_1)$

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**Algorithm 3** Reformulated overlapping multiplicative Schwarz

---

Given  $u^{\{0\}}$  defined on  $\Omega$

**for**  $k = 0, 1, 2, \dots, K - 1$  **do**

**for**  $i = 1, 2, \dots, p$  **do**

        Find  $w^{\{k+i/p\}} \in H_0^1(\Omega_1)$  such that

$$a_i \left( w^{\{k+i/p\}}, v \right) = l_i(v) - a_i \left( u^{\{k+(i-1)/p\}}, v \right), \quad \forall v \in H_0^1(\Omega_i). \quad (3)$$

        Let

$$u^{\{k+i/p\}} = \begin{cases} u^{\{k+(i-1)/p\}} + w^{\{k+i/p\}}, & \text{on } \bar{\Omega}_i, \\ u^{\{k+(i-1)/p\}}, & \text{on } \Omega \setminus \bar{\Omega}_i. \end{cases} \quad (4)$$

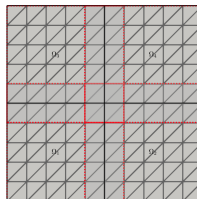
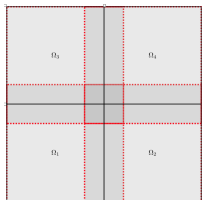
**end for**

**end for**

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# Finite element discretization

- ▶ Construct a triangulation of  $\Omega_i$ .
- ▶ Substitute finite dimensional spaces  $V_{i,h}^k \subset H_{D^k}^1(\Omega_i)$  and  $V_{i,h,0} \subset H_0^1(\Omega_i)$ .
- ▶ Discrete solutions  $U^{\{k+i/p\}} \in V_h$  and in the reformulated case  $W^{\{k+i/p\}} \in V_{i,h,0} \subset H_0^1(\Omega_i)$ .



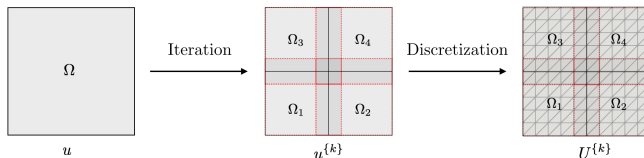
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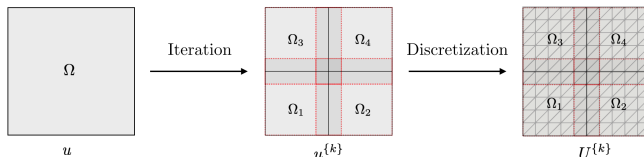
Finite iteration FEM solution,  $U^{\{k\}}$ , is the discretization this new operator

# Discretization and iteration errors

QoI

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Decompose error

$$(u - U^{\{k\}}, \psi) = \underbrace{(u - u^{\{k\}}, \psi)}_{\text{Iteration Error}} + \underbrace{(u^{\{k\}} - U^{\{k\}}, \psi)}_{\text{Discretization Error}} = e_I^{\{k\}} + e_D^{\{k\}}.$$

## Analogy to linear algebra

1. One-way coupling problems ( $Kp$  of them)
2. Gauss-Seidel iteration

Solve

$$Mx = b$$

where  $M$  is a  $p \times p$  matrix using  $K$  Gauss-Seidel iterations.

(Here we choose  $p = 4$  and  $K = 5$ .)

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Decompose  $M$  as a sum of strictly lower triangular, diagonal and a strictly upper triangular matrices

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Decompose  $M$  as a sum of strictly lower triangular, diagonal and a strictly upper triangular matrices

$$M = L + D + U$$

Solve

$$(L + D)x^{\{k+1\}} = b - Ux^{\{k\}}, k = 0, 1, \dots$$

Let  $A = (L + D)$  and  $B = U$ , each of which are  $p \times p$  matrices.

# Gauss-Seidel iteration in block form

The complete Gauss-Seidel iteration can be written as the following block lower triangular system

$$C_{\text{gs}}x = b,$$

where

$$C_{\text{gs}} = \begin{bmatrix} \mathbf{A} & 0 & 0 & 0 & 0 \\ \mathbf{B} & \mathbf{A} & 0 & 0 & 0 \\ 0 & \mathbf{B} & \mathbf{A} & 0 & 0 \\ 0 & 0 & \mathbf{B} & \mathbf{A} & 0 \\ 0 & 0 & 0 & \mathbf{B} & \mathbf{A} \end{bmatrix}.$$

$C_{\text{gs}}$  has a block bandwidth of two. Recall  $\mathbf{A}$  is lower triangular and  $\mathbf{B}$  is strictly upper triangular. Here

$$\mathbf{x}^\top = \left( x^{\{1\}\top}, x^{\{2\}\top}, x^{\{3\}\top}, x^{\{4\}\top}, x^{\{5\}\top} \right)$$

and

$$\mathbf{b}^\top = \left( b^\top, b^\top, b^\top, b^\top, b^\top \right).$$

The adjoint problem is  $C_{\text{gs}}^\top \phi = \psi$  where

$$\psi^\top = (0, 0, 0, 0, \psi^\top)^\top, \quad \phi^\top = (\phi^{[1]\top}, \phi^{[2]\top}, \phi^{[3]\top}, \phi^{[4]\top}, \phi^{[5]\top})^\top.$$

and

$$C_{\text{gs}}^\top = \begin{bmatrix} \mathbf{A}^\top & \mathbf{B}^\top & 0 & 0 & 0 \\ 0 & \mathbf{A}^\top & \mathbf{B}^\top & 0 & 0 \\ 0 & 0 & \mathbf{A}^\top & \mathbf{B}^\top & 0 \\ 0 & 0 & 0 & \mathbf{A}^\top & \mathbf{B}^\top \\ 0 & 0 & 0 & 0 & \mathbf{A}^\top \end{bmatrix}.$$

The adjoint problems are

$$\mathbf{A}^\top \phi^{[5]} = \psi,$$

$$\mathbf{A}^\top \phi^{[4]} = -\mathbf{B}^\top \phi^{[4]}, \quad \mathbf{A}^\top \phi^{[3]} = -\mathbf{B}^\top \phi^{[3]}, \quad \mathbf{A}^\top \phi^{[2]} = -\mathbf{B}^\top \phi^{[2]}, \quad \mathbf{A}^\top \phi^{[1]} = -\mathbf{B}^\top \phi^{[1]}$$

which can be solved sequentially by backward substitution.

# Resolving the Gauss-Seidel iteration blocks

Note that

$$C_{gs} = \left[ \begin{array}{cccc|cccc|c} A_{11} & & & & & & & & \\ A_{21} & A_{22} & & & & & & & \\ A_{31} & A_{32} & A_{33} & & & & & & \\ A_{41} & A_{42} & A_{43} & A_{44} & & & & & \\ \hline & A_{12} & A_{13} & A_{14} & A_{11} & & & & \\ & & A_{23} & A_{24} & A_{21} & A_{22} & & & \\ & & & A_{34} & A_{31} & A_{32} & A_{33} & & \\ & & & & A_{41} & A_{42} & A_{43} & A_{44} & \\ \hline & & & & & & & \vdots & \\ & & & & & & & & \dots \end{array} \right].$$

Now  $x^{\{k\}} = [x_1^{\{k\}}, x_2^{\{k\}}, x_3^{\{k\}}, x_4^{\{k\}}]$

# Resolving the Gauss-Seidel iteration adjoint blocks

Hence (allowing the possibility for  $A_{ij}$  to themselves be matrix-valued),

$$C_{gs}^{\top} = \left[ \begin{array}{cccc|cccc|c} A_{11}^{\top} & A_{21}^{\top} & A_{31}^{\top} & A_{41}^{\top} & & & & & \\ & A_{22}^{\top} & A_{32}^{\top} & A_{42}^{\top} & A_{12}^{\top} & & & & \\ & & A_{33}^{\top} & A_{43}^{\top} & A_{13}^{\top} & A_{23}^{\top} & & & \\ & & & A_{44}^{\top} & A_{14}^{\top} & A_{24}^{\top} & A_{34}^{\top} & & \\ \hline & & & & A_{11}^{\top} & A_{21}^{\top} & A_{31}^{\top} & A_{41}^{\top} & \\ & & & & & A_{22}^{\top} & A_{32}^{\top} & A_{42}^{\top} & \\ & & & & & & A_{33}^{\top} & A_{43}^{\top} & \\ & & & & & & & A_{44}^{\top} & \\ \hline & & & & & & & \vdots & \\ & & & & & & & & \dots \end{array} \right].$$

Now  $\phi^{[k]} = [\phi_1^{[k]}, \phi_2^{[k]}, \phi_3^{[k]}, \phi_4^{[k]}]$

The adjoint equations within each individual block can also be solved via backward substitution.

The adjoint equations are therefore

$$A_{ii}^\top \phi_i^{[K]} = \psi_i - \sum_{j>i} A_{ji}^\top \phi_j^{[K]}, \quad i = 1, \dots, p,$$

$$A_{ii}^\top \phi_i^{[k]} = - \sum_{j<i} A_{ji}^\top \phi_j^{[k+1]} - \sum_{j>i} A_{ji}^\top \phi_j^{[k]}, \quad i = 1, \dots, p, \quad k = K-1, \dots, 1.$$

# Solving the Gauss-Seidel adjoint by back substitution

The adjoint equations are therefore

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$$A_{ii}^\top \phi_i^{[k]} = - \sum_{j<i} A_{ji}^\top \phi_j^{[k+1]} - \sum_{j>i} A_{ji}^\top \phi_j^{[k]}, \quad i = 1, \dots, p, \quad k = K-1, \dots, 1.$$

Since  $C_{\text{gs}}\mathbf{x} = \mathbf{b}$  is just a  $Kp \times Kp$  linear system,

$$(\mathbf{e}, \psi) = (\mathcal{R}, \phi)$$

where

$$\mathcal{R} = \mathbf{b} - C_{\text{gs}}\hat{\mathbf{x}}$$

for approximate solution  $\hat{\mathbf{x}}$ .

The error in the quantity of interest can be expressed as an inner products of two vectors of length  $Kp$ .

**Partition of Unity** subject to cover  $\Omega_i$

$$\sum_{i=1}^p \chi_i(x) = 1, \quad x \in \Omega, \quad \chi_i = 0, \quad x \in \Omega \setminus \Omega_i$$

Define **local QoI data**

$$\psi_i = \chi_i \psi.$$

This leads to

$$(e_D^{\{k\}}, \psi) = (e_D^{\{k\}}, \sum_{i=1}^p \chi_i \psi) = \sum_{i=1}^p (e_D^{\{k\}}, \psi_i)_i.$$

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**Weak residuals** for  $i = 1, 2, \dots, p$ :

$$R_i(s, v) = l_i(v) - a_i(s, v).$$

Define adjoints  $\phi^{[k+i/p]}$  belonging to  $V_i$  as follows:

$$a_i(v, \phi^{[K-1+i/p]}) = \sum_{j=1}^p (\psi_j, v)_{ij} - \sum_{j=i+1}^p a_{ij}(v, \phi^{[K-1+j/p]})$$

$$\forall v \in V_i,$$

$$a_i(v, \phi^{[k+i/p]}) = - \sum_{j=i+1}^p a_{ij}(v, \phi^{[k+j/p]}) - \sum_{j=1}^{i-1} a_{ij}(v, \phi^{[k+1+j/p]})$$

$$\forall v \in V_i, \quad 0 \leq k < K-1.$$

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 &\quad \forall v \in V_i, \\
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 &\quad \forall v \in V_i, \quad 0 \leq k < K-1.
 \end{aligned}$$

Theorem (Discretization error for multiplicative Schwarz)

$$(\psi, u^{\{K\}} - U^{\{K\}}) = \sum_{k=0}^{K-1} \sum_{i=1}^p R_i(\tilde{U}^{\{k+i/p\}}, \phi^{[k+i/p]})$$

## Gauss-Seidel

$$A_{ii}^\top \phi_i^{[K]} = \psi_i - \sum_{j>i} A_{ji}^\top \phi_j^{[K]}$$

$$A_{ii}^\top \phi_i^{[k]} = - \sum_{j<i} A_{ji}^\top \phi_j^{[k+1]} - \sum_{j>i} A_{ji}^\top \phi_j^{[k]}$$

## Error in Gauss-Seidel

$$(\mathbf{e}, \boldsymbol{\psi}) = (\mathcal{R}, \boldsymbol{\phi})$$

## Multiplicative Schwarz

$$\begin{aligned} a_i(v, \phi^{[K-1+i/p]}) &= \sum_{j=1}^p (\psi_j, v)_{ij} \\ &\quad - \sum_{j=i+1}^p a_{ij}(v, \phi^{[K-1+j/p]}) \end{aligned}$$

$$\begin{aligned} a_i(v, \phi^{[k+i/p]}) &= - \sum_{j=i+1}^p a_{ij}(v, \phi^{[k+j/p]}) \\ &\quad - \sum_{j=1}^{i-1} a_{ij}(v, \phi^{[k+1+j/p]}) \end{aligned}$$

## Discretization error in multiplicative Schwarz

$$(\psi, u^{\{K\}} - U^{\{K\}}) = \sum_{k=0}^{K-1} \sum_{i=1}^p R_i(\tilde{U}^{\{k+i/p\}}, \phi^{[k+i/p]})$$

# Analysis with overlapping subdomains

Overlap between subdomains complicates the analysis.

The error information has to be tracked through the overlaps and across iterations.

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Let  $e_W^{\{k\}} = w^{\{k\}} - W^{\{k\}}$ .

Lemma 1 [Error in QoI in terms of discretization errors with homogeneous bcs]

The discretization error in the QoI is

$$\left(e_D^{\{K\}}, \psi\right) = \sum_{k=0}^{K-1} \sum_{i=1}^p \sum_{j=1}^p \left(e_W^{\{k+i/p\}}, \psi_j\right)_{ij}.$$

# Overlapping domains complicate the analysis

## Lemma 2 [Bilinear form with discretization errors with homogeneous bcs]

$$a_i \left( e_W^{\{i/p\}}, v \right) = a_i \left( e_D^{\{i/p\}}, v \right) - \sum_{r=1}^{i-1} a_{ir} \left( e_W^{\{r/p\}}, v \right),$$

and for  $k \geq 1$ ,

$$\begin{aligned} a_i \left( e_W^{\{k+i/p\}}, v \right) &= a_i \left( e_D^{\{k+i/p\}}, v \right) - a_i \left( e_D^{\{k-1+i/p\}}, v \right) \\ &\quad - \sum_{r=1}^{i-1} a_{ir} \left( e_W^{\{k+r/p\}}, v \right) - \sum_{r=i+1}^p a_{ir} \left( e_W^{\{k-1+r/p\}}, v \right). \end{aligned}$$

## Lemma 3 [Sums of bilinear form with discretization errors with homogeneous bcs]

For  $0 \leq Q \leq K-1$  we have

$$\begin{aligned} \sum_{k=0}^Q a_i \left( e_W^{\{k+i/p\}}, v \right) &= a_i \left( e_D^{\{Q+i/p\}}, v \right) - \sum_{k=0}^Q \sum_{r=1}^{i-1} a_{ir} \left( e_W^{\{k+r/p\}}, v \right) \\ &\quad - \sum_{k=0}^{Q-1} \sum_{r=i+1}^p a_{ir} \left( e_W^{\{k+r/p\}}, v \right). \end{aligned}$$

# Overlapping domains complicate the analysis

Simplify the adjoint right hand side

$$\tau_i^Q(v) = \begin{cases} \sum_{j=1}^p (v, \psi_j)_{ij}, & Q = K - 1, \\ - \sum_{j=1}^{i-1} a_{ij} \left( v, \phi^{[Q+1+j/p]} \right), & 0 \leq Q < K - 1. \end{cases}$$

Lemma 4 [Sum of RHS of the adjoint equations over iterations]

Let  $2 \leq M \leq p + 1$  and  $R = M - 1$  and  $0 \leq Q < K$ . Then

$$\begin{aligned} & \sum_{k=0}^Q \tau_R^Q \left( e_W^{\{k+R/p\}} \right) - \sum_{k=0}^Q \sum_{j=M}^p a_{Rj} \left( e_W^{\{k+R/p\}}, \phi^{[Q+j/p]} \right) \\ &= a_R \left( e_D^{\{Q+R/p\}}, \phi^{[Q+R/p]} \right) - \sum_{k=0}^Q \sum_{i=1}^{R-1} a_{iR} \left( e_W^{\{k+i/p\}}, \phi^{[Q+R/p]} \right) \\ & \quad - \sum_{k=0}^{Q-1} \sum_{i=M}^p a_{iR} \left( e_W^{\{k+i/p\}}, \phi^{[Q+R/p]} \right). \end{aligned}$$

# Overlapping domains complicate the analysis

## Lemma 5 [Sum of RHS of the adjoint equations over iterations and subdomains]

Let  $1 \leq M \leq p + 1$  and  $0 \leq Q < K$ . Then,

$$\begin{aligned} I = \sum_{k=0}^Q \sum_{i=1}^p \tau_i^Q \left( e_W^{\{k+i/p\}} \right) &= \sum_{i=M}^p a_i \left( e_D^{\{Q+i/p\}}, \phi^{[Q+i/p]} \right) + \sum_{k=0}^Q \sum_{i=1}^{M-1} \tau_i^Q \left( e_W^{\{k+i/p\}} \right) \\ &\quad - \sum_{k=0}^Q \sum_{i=1}^{M-1} \sum_{j=M}^p a_{ij} \left( e_W^{\{k+i/p\}}, \phi^{[Q+j/p]} \right) \\ &\quad - \sum_{k=0}^{Q-1} \sum_{i=M+1}^p \sum_{j=M}^{i-1} a_{ij} \left( e_W^{\{k+i/p\}}, \phi^{[Q+j/p]} \right). \end{aligned}$$

## Corollary 1

Let  $0 \leq Q < K$ . Then we have

$$\sum_{k=0}^Q \sum_{i=1}^p \tau_i^Q \left( e_W^{\{k+i/p\}} \right) = \sum_{i=1}^p a_i \left( e_D^{\{Q+i/p\}}, \phi^{[Q+i/p]} \right) + \sum_{k=0}^{Q-1} \sum_{i=1}^p \tau_i^{Q-1} \left( e_W^{\{k+i/p\}} \right).$$

Define the *global* adjoint

$$a(v, \phi) = (\psi, v) \quad \forall v \in H_0^1(\Omega).$$

Theorem (Total error representation )

$$(u - U^{\{K\}}, \psi) = R(U^{\{K\}}, \phi)$$

The *iteration* error is given by the *difference* of the total and discretization errors.

Theorem (Iteration error representation)

$$(u - u^{\{K\}}, \psi) = R(U^{\{K\}}, \phi) - (\psi, u^{\{K\}} - U^{\{K\}})$$

# Error Estimates and Effectivity Ratios

Let  $\Phi^{[k+i/p]} \approx \phi^{[k+i/p]}$  and  $\Phi \approx \phi$ .

Error estimate for the discretization error

$$\eta_D^K \equiv \sum_{k=0}^{K-1} \sum_{i=1}^p R_i(\tilde{U}^{\{k+i/p\}}, \Phi^{[k+i/p]}).$$

Effectivity ratio for discretization error is

$$\gamma_D = \frac{\eta_D^K}{(u^{\{K\}} - U^{\{K\}}, \psi)}.$$

Error estimate for total error

$$\eta^K \equiv R(U^{\{K\}}, \Phi).$$

Effectivity ratio for total error is

$$\gamma = \frac{\eta^{\{K\}}}{(u - U^{\{K\}}, \psi)}.$$

# Estimates for the Poisson's Equation

Consider the Poisson's Equation

$$\begin{aligned} -\nabla \cdot \nabla u &= f & \text{in } \Omega \\ u &= 0 & \text{on } \partial\Omega \end{aligned}$$

where  $\Omega = [0.0, 1.0] \times [0.0, 1.0]$ ,  $f(x, y) = 8\pi^2 \sin(2\pi x) \sin(2\pi y)$ .

The adjoint data is

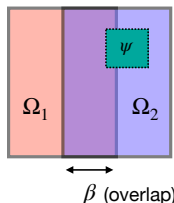
$$\psi = \chi_{[0.6, 0.8] \times [0.6, 0.8]}$$

**Mesh**  $2 \times N_x \times N_y$  linear triangular elements.

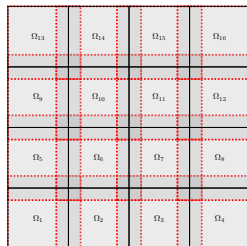
**Primal problems** solved using continuous piecewise linear elements.

**Adjoint problems** solved using continuous piecewise quadratic elements.

(Implementation using Fenics.)



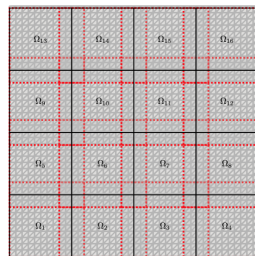
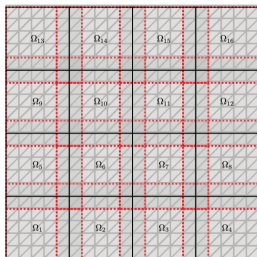
# MS: Poisson's equation – $4 \times 4$ domains. Varying number of iterations



| $K$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-----|----------------|----------|------------------|------------|------------------|
| 2   | 9.22e-03       | 1.00e+00 | 1.02e-03         | 1.00e+00   | 8.20e-03         |
| 4   | 2.72e-03       | 9.99e-01 | 8.27e-04         | 9.99e-01   | 1.90e-03         |

Table:  $\beta = 0.1$ ,  $N_x = 20$ ,  $N_y = 20$ ,

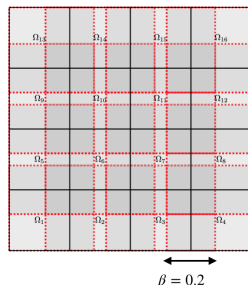
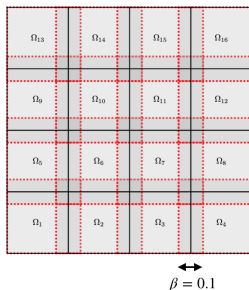
# MS: Poisson's equation – $4 \times 4$ domains. Refining the mesh



| $N_x$ | $N_y$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-------|-------|----------------|----------|------------------|------------|------------------|
| 20    | 20    | 9.22e-03       | 1.00e+00 | 1.02e-03         | 1.00e+00   | 8.20e-03         |
| 40    | 40    | 8.45e-03       | 1.00e+00 | 2.55e-04         | 1.00e+00   | 8.20e-03         |

Table:  $K = 2$ ,  $\beta = 0.1$ ,

# MS: Poisson's equation – $4 \times 4$ domains. Changing the overlap



| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.1     | 9.22e-03       | 1.00e+00 | 1.02e-03         | 1.00e+00   | 8.20e-03         |
| 0.2     | 2.80e-03       | 9.99e-01 | 7.31e-04         | 9.98e-01   | 2.07e-03         |

Table:  $K = 2$ ,  $N_x = 20$ ,  $N_y = 20$ ,

## MS: Poisson's equation – $2 \times 1$ domains. Cancellation of error

Here

$$\psi = \chi_{[0.4, 0.8] \times [0.4, 0.8]}$$

Note the cancellation of error between the discretization and iteration errors when  $K = 6$ .

| $K$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-----|----------------|----------|------------------|------------|------------------|
| 1   | 3.98e-03       | 1.00e+00 | -1.50e-05        | 9.98e-01   | 4.00e-03         |
| 2   | 2.07e-03       | 1.00e+00 | -5.95e-05        | 9.99e-01   | 2.13e-03         |
| 3   | 1.04e-03       | 1.00e+00 | -9.11e-05        | 1.00e+00   | 1.13e-03         |
| 4   | 4.89e-04       | 1.00e+00 | -1.14e-04        | 1.00e+00   | 6.03e-04         |
| 5   | 1.91e-04       | 1.00e+00 | -1.30e-04        | 1.00e+00   | 3.21e-04         |
| 6   | 2.97e-05       | 1.01e+00 | -1.41e-04        | 1.00e+00   | 1.71e-04         |
| 7   | -5.83e-05      | 9.96e-01 | -1.49e-04        | 1.00e+00   | 9.09e-05         |
| 8   | -1.07e-04      | 9.98e-01 | -1.55e-04        | 1.00e+00   | 4.84e-05         |
| 9   | -1.33e-04      | 9.98e-01 | -1.59e-04        | 1.00e+00   | 2.58e-05         |
| 10  | -1.48e-04      | 9.98e-01 | -1.62e-04        | 1.00e+00   | 1.37e-05         |

Table:  $\beta = 0.05$ ,  $N_x = 40$ ,  $N_y = 40$

Consider the convection-diffusion equation

$$\begin{aligned} -\nabla \cdot \nabla u + \mathbf{b} \cdot \nabla u &= f && \text{in } \Omega \\ u &= 0 && \text{on } \partial\Omega \end{aligned}$$

where  $\Omega = [0.0, 1.0] \times [0.0, 1.0]$ ,  $f(x, y) = 1$ , and  $\mathbf{b} = [-60, 0]$ . The effect of this convective vector field is to carry information from right to left. Let

$$\psi = \chi_{[0.05, 0.2] \times [0.05, 0.2]},$$

chosen as there are sharp gradients in the solution near the left boundary due to the convective vector field  $\mathbf{b}$ .

(The adjoint was computed using continuous piecewise cubic polynomials.)

## A $4 \times 1$ configuration

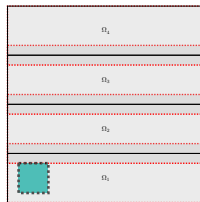
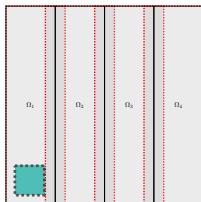
| $K$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-----|----------------|----------|------------------|------------|------------------|
| 2   | 9.76e-03       | 1.00e+00 | -1.54e-04        | 9.87e-01   | 9.92e-03         |

Table:  $\beta = 0.1$ ,  $N_x = 20$ ,  $N_y = 20$ ,

## B $1 \times 4$ configuration

| $K$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-----|----------------|----------|------------------|------------|------------------|
| 2   | 8.42e-05       | 1.03e+00 | -3.73e-04        | 9.94e-01   | 4.57e-04         |

Table:  $\beta = 0.1$ ,  $N_x = 20$ ,  $N_y = 20$ ,



Define

$$S_i^K = \sum_{k=0}^{K-1} R_i(\tilde{U}^{\{k+i/p\}}, \phi^{[k+i/p]})$$

where  $S_i^K$  indicates the discretization error contribution of subdomain  $i$ , i.e.,

$$(\psi, u^{\{K\}} - U^{\{K\}}) = \sum_{i=1}^p S_i^K$$

## MS: Two-stage solution strategy (discretization error dominant)

Poisson's equation – Using the estimates on  $2 \times 2$  subdomains.

| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.2     | 2.36e-03       | 9.83e-01 | 2.36e-03         | 9.89e-01   | 6.98e-06         |

| $i$     | 1        | 2         | 3         | 4        |
|---------|----------|-----------|-----------|----------|
| $S_i^K$ | 3.07e-04 | -7.94e-04 | -7.82e-04 | 3.62e-03 |

Table:  $K = 6$ , Number of vertices = 121

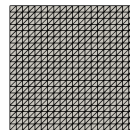
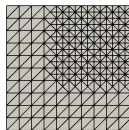
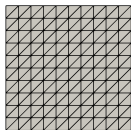
Subdomain 4 contributes the most towards the discretization error. Moreover, it has opposite sign to those of the next two largest subdomain contributions.

## MS: Two-stage solution strategy (discretization error dominant)

| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.2     | 3.44e-04       | 1.00e+00 | 3.37e-04         | 1.05e+00   | 6.97e-06         |

| $i$     | 1        | 2         | 3         | 4        |
|---------|----------|-----------|-----------|----------|
| $S_i^K$ | 1.82e-04 | -3.87e-04 | -3.85e-04 | 9.27e-04 |

Table:  $K = 6$ , Number of vertices = 253



Error on uniformly refined mesh with 441 vertices is  $6.24 \times 10^{-4}$ !

## MS: Two-stage solution strategy (iteration error dominant)

Poisson's equation – Using the estimates on  $2 \times 2$  subdomains.

| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.05    | 1.23e-03       | 1.00e+00 | 1.79e-04         | 9.99e-01   | 1.05e-03         |

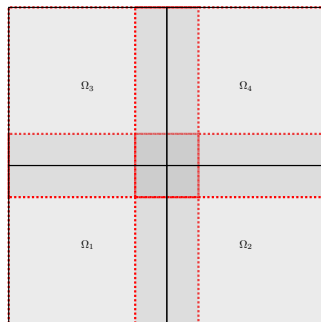
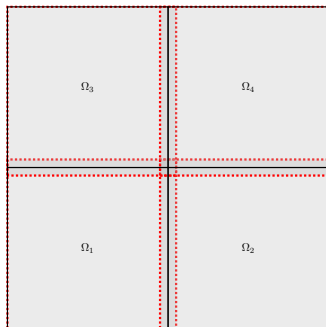
Table:  $K = 2$ ,  $N_x = 40$ ,  $N_y = 40$ ,

The iteration error is the dominant error. There are two ways to reduce it, either by more iterations or increasing  $\beta$ . We choose the latter.

## MS: Two-stage solution strategy (iteration error dominant)

| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.2     | 5.04e-04       | 1.00e+00 | 1.62e-04         | 9.99e-01   | 3.42e-04         |

Table:  $K = 2$ ,  $N_x = 40$ ,  $N_y = 40$ ,



# Additive Schwarz: Algorithm

Here  $\tau$  is the Richardson parameter needed to ensure convergence.

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**Algorithm 4** Overlapping additive Schwarz

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Given  $u^{\{0\}}$  defined on  $\Omega$ .

**for**  $k = 0, 1, 2, \dots, K - 1$  **do**

**for**  $i = 1, 2, \dots, p$  **do**

        Find  $\tilde{u}^{\{k+i/p\}} \in H_{D_k}^1(\Omega_i)$  such that

$$a_i(\tilde{u}_i^{\{k+1\}}, v) = l_i(v) \quad \forall v \in H_0^1(\Omega_i).$$

$$\text{Let } E\tilde{u}_i^{\{k+1\}} = \begin{cases} \tilde{u}_i^{\{k+1\}} & \text{on } \overline{\Omega}_i \\ u^{\{k\}} & \text{on } \Omega \setminus \overline{\Omega}_i. \end{cases}$$

Let

$$u^{\{k+1\}} = (1 - \tau p)u^{\{k\}} + \tau \left( \sum_{i=1}^p E\tilde{u}_i^{\{k+1\}} \right).$$

**end for**

**end for**

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**Algorithm 5** Reformulated overlapping additive Schwarz

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Given  $u^{\{0\}}$  defined on  $\Omega$ .

**for**  $k = 0, 1, 2, \dots, K - 1$  **do**

**for**  $i = 1, 2, \dots, p$  **do**

        Find  $w_i^{\{k+1\}} \in H_0^1(\Omega_i)$  such that

$$a_i(w_i^{\{k+1\}}, v) = l_i(v) - a_i(u^{\{k\}}, v) \quad \forall v \in H_0^1(\Omega_i).$$

$$\text{Let } \tilde{E}w_i^{\{k+1\}} = \begin{cases} w_i^{\{k+1\}} & \text{on } \overline{\Omega}_i \\ 0 & \text{on } \Omega \setminus \overline{\Omega}_i. \end{cases}$$

Let

$$u^{\{k+1\}} = u^{\{k\}} + \tau \left( \sum_{i=1}^p \tilde{E}w_i^{\{k+1\}} \right).$$

**end for**

**end for**

---

## Adjoint problems

$$a_i(v, \phi_i^{[k]}) = \tau \sum_{j=1}^p \left[ (\psi_j, v)_{ij} - a_{ij} \left( v, \sum_{l=k+1}^K \phi_j^{[l]} \right) \right].$$

## Theorem (Discretization error representation for additive Schwarz)

$$(\psi, u^{\{K\}} - U^{\{K\}}) = \sum_{k=1}^K \sum_{i=1}^p R_i(\tilde{U}^{\{k\}}, \phi^{[k]})$$

# AS: Poisson's equation – $4 \times 4$ domains

| $\beta$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|---------|----------------|----------|------------------|------------|------------------|
| 0.1     | 2.18e-02       | 1.00e+00 | 6.83e-04         | 1.00e+00   | 2.12e-02         |
| 0.2     | 1.25e-02       | 1.00e+00 | 6.14e-04         | 1.00e+00   | 1.18e-02         |

Table:  $K = 2$ ,  $N_x = 20$ ,  $N_y = 20$ ,

| $K$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-----|----------------|----------|------------------|------------|------------------|
| 2   | 2.18e-02       | 1.00e+00 | 6.83e-04         | 1.00e+00   | 2.12e-02         |
| 4   | 1.58e-02       | 1.00e+00 | 9.42e-04         | 9.86e-01   | 1.48e-02         |

Table:  $\beta = 0.1$ ,  $N_x = 20$ ,  $N_y = 20$ ,

| $N_x$ | $N_y$ | $\eta^{\{K\}}$ | $\gamma$ | $\eta_D^{\{K\}}$ | $\gamma_D$ | $\eta_I^{\{K\}}$ |
|-------|-------|----------------|----------|------------------|------------|------------------|
| 20    | 20    | 2.18e-02       | 1.00e+00 | 6.83e-04         | 1.00e+00   | 2.12e-02         |
| 40    | 40    | 2.13e-02       | 1.00e+00 | 1.70e-04         | 1.00e+00   | 2.12e-02         |

Table:  $K = 2$ ,  $\beta = 0.1$ ,

1. Optimized Schwarz (Robin boundary conditions on subdomain interfaces)
2. Restricted Additive Schwarz (default preconditioner in PETSc)
3. Asynchronous variants
4. Time-dependent problems
5. Coupling with parallel-in-time integration (combining with previous analysis for the parareal method)

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